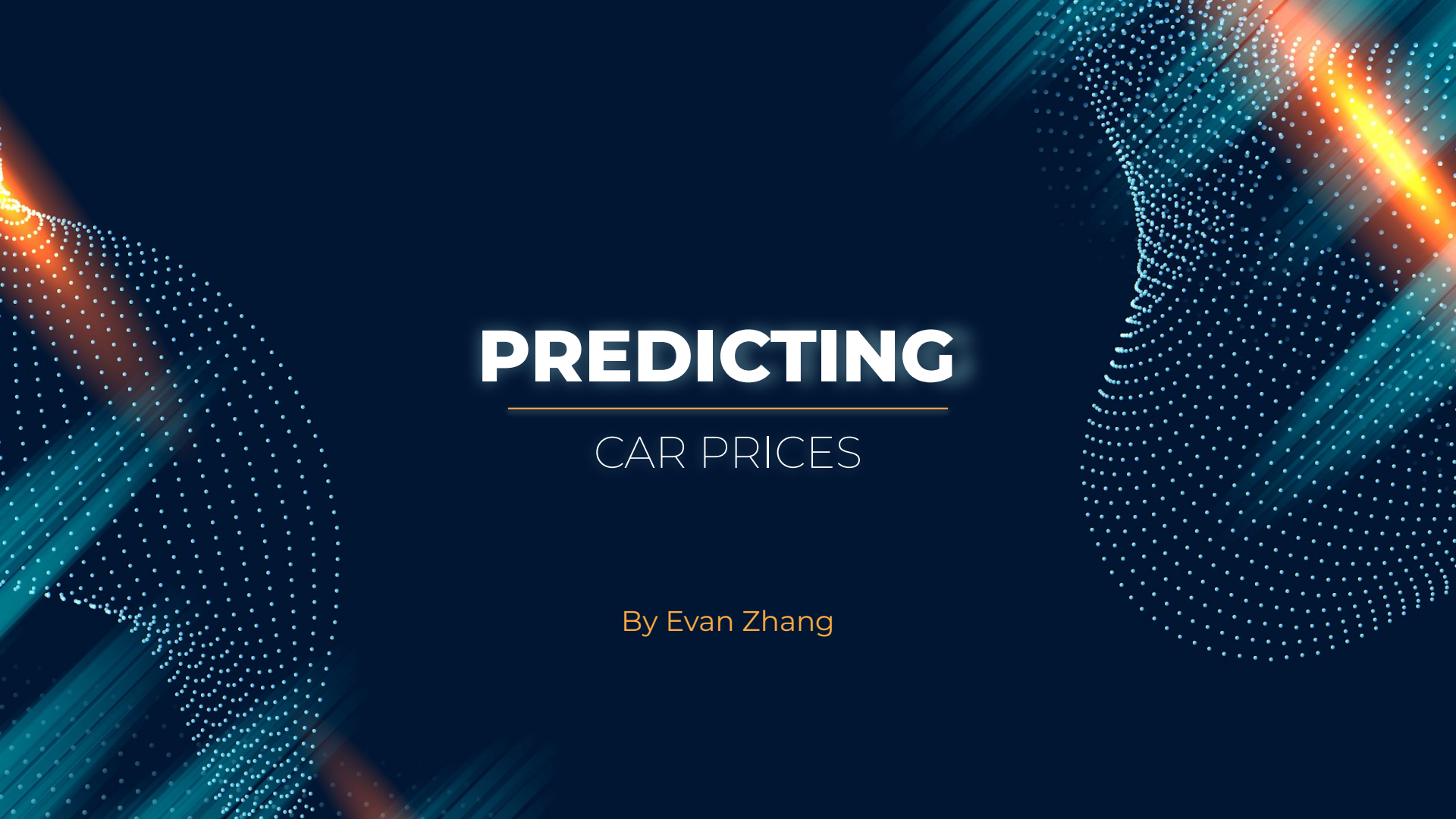




PREDICTING

CAR PRICES

By Evan Zhang

MAIN QUESTION:

CAN A MACHINE LEARNING MODEL
ACCURATELY PREDICT THE PRICE OF A CAR?



MY DATASET

I found my data set from the UC Irvine Machine Learning Repository, the dataset is called [Automobile](#)

The dataset has 25 features some are:

Manufacturers, model, horsepower, compression ratio, engine size, number of cylinders, etc.

SOME PROBLEMS:

- The dataset was made in 1985 so very new car prices or very expensive cars will not be accurately but normal modern cars under 100,000 dollars will still be accurately predicted.
- The dataset that I found had their target variable was symboling, not the prices so I had to change that.

HOW I TRAINED IT

I trained the model with linear regression and hyperparameters and keras tuner so its like better.

HERE IS THE UI THAT I MADE

Automobile Price Valuation Estimator

Performance & Engine Specs

Engine Size (cid) 

60 330

☐  Don't know Engine Size (Uses average)

Horsepower 

40 800

☐  Don't know Horsepower (Uses average)

Peak RPM 

4000 7000

☐  Don't know Peak RPM (Uses average)

Compression Ratio

☐  Don't know Compression Ratio (Uses average)

Bore

☐  Don't know Bore (Uses average)

Stroke

☐  Don't know Stroke (Uses average)

Curb Weight (lbs) 

1400 4100

☐  Don't know Curb Weight (Uses average)

Fuel System

Physical Dimensions & Logistics

Manufacturer (Make)

Body Style

Drive Wheels ☐ Don't Know ☒ fwd ☐ rwd ☐ 4wd

Engine Location ☐ Don't Know ☒ front ☐ rear

Number of Doors ☐ Don't Know ☐ two ☒ four

Length (inches)

☐  Don't know Length (Uses average)

Width (inches)

☐  Don't know Width (Uses average)

Height (inches)

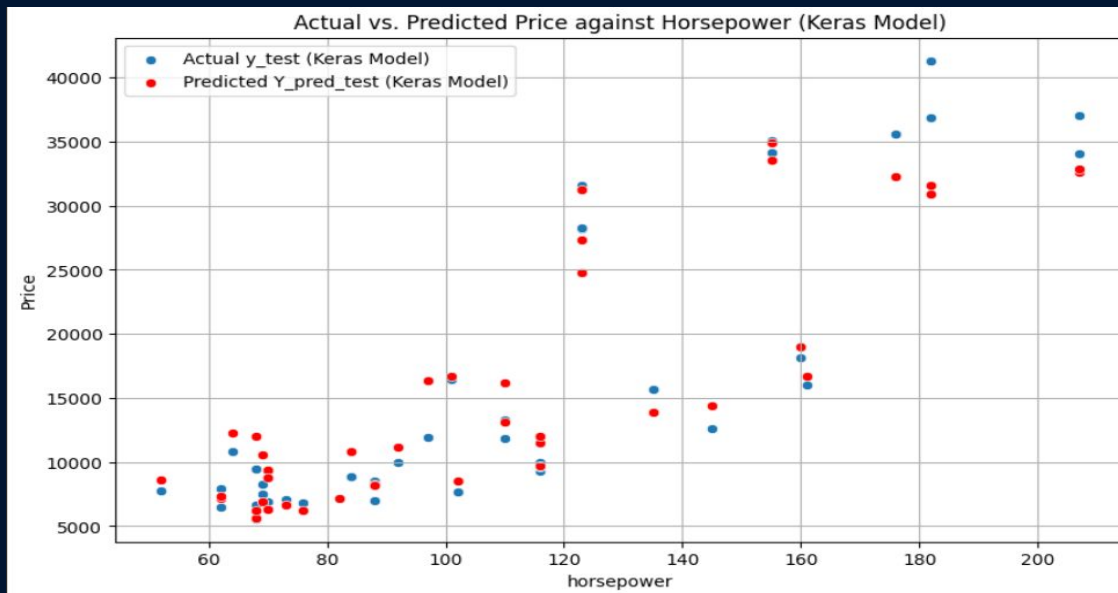
☐  Don't know Height (Uses average)

Wheel Base (inches)

☐  Don't know Wheel Base (Uses average)

RESULTS:

At the end, I got an mse of 4,672,860.32 but it isn't actually that bad because the mae was 1,387.27 which means that the price was about 3387 dollars off.



EXAMPLE

--- Other Specs ---

Determin

Estimated Market Value

\$76,700.00 (Demo Mode - Model Not Loaded)

What is the price of Mercedes GL 500?

AI Overview

The Mercedes-Benz GL 500 (sold in the US as the GL 550) is out of production, with models typically ranging from **\$10,000 to \$40,000** depending on the year and model, whereas its original new starting price was roughly **\$80,000 to \$91,000**. [Kelley Blue Book](#)

Used Market and Original Pricing

- **Older Models (2007–2012 GL 500 / GL 550):** Used prices range from **\$10,000 to \$18,000** from an original MSRP of about **\$79,360 to \$85,300**. [Mercedes-Benz USA +2](#)
- **Newer Generation (2013–2016 GL 550):** Used prices range from **\$25,000 to \$41,000**, from an original MSRP starting around **\$86,900 to \$91,300**. [Mercedes-Benz USA +2](#)
- **International GL 500 Markets:** In regions like Australia or Europe where the "GL 500" badge is explicitly used, late models run upwards of **\$30,000 to \$36,000 USD equivalent (€29,900 to \$35,990 AUD)**. [Carsales +1](#)

IF I COULD DO IT AGAIN

1. I would find a newer dataset so the model would be better at predicting more modern cars.
2. I would use a different website hoster so the website would not delete itself in a week.



THANK YOU

From Evan Zhang